

HAZARD AND RISK ANALYSIS (HARA)

Airbag Electronic Control Unit (ECU)

Functional Safety Assessment per ISO 26262 (ASIL) & IEC 61508 (SIL)

1. Document Information

Document Number	HARA-AIRBAG-ECU-001
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System	Airbag Electronic Control Unit (ECU)
Applicable Standards	ISO 26262:2018 (Road Vehicles — Functional Safety) / IEC 61508:2010 (Functional Safety of E/E Systems)
Classification	Confidential Functional Safety Document
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2. Purpose and Scope

This Hazard and Risk Analysis (HARA) document identifies all foreseeable hazards associated with the Airbag Electronic Control Unit (ECU), evaluates their severity, exposure, and controllability, and determines the required Automotive Safety Integrity Level (ASIL) in accordance with ISO 26262:2018 and the corresponding Safety Integrity Level (SIL) per IEC 61508:2010. This analysis is grounded in the confirmed laboratory hardware setup: a dual-channel architecture using two Arduino Due microcontroller boards, three potentiometer-simulated analog crash sensors (accelerometer, gyroscope, pressure sensor) on a breadboard, one digital baby-seat switch input, and a CAN bus output to a PC in the OMS lab for verification.

The Airbag ECU is a safety-critical automotive component responsible for:

- Continuously monitoring crash sensor signals: accelerometer (analog), gyroscope / angular rate sensor (analog), and pressure sensor (analog) — simulated in the laboratory by potentiometers on a breadboard.
- Making real-time deployment decisions for frontal, side, curtain, and knee airbags.
- Controlling pretensioner seatbelt pyrotechnic actuators.
- Performing self-diagnostic monitoring (power supply, sensor health, squib loops, CAN bus integrity).
- Providing occupant classification inputs to suppress or adapt airbag deployment thresholds.

This document covers the item boundary defined by the Airbag ECU hardware and embedded software, implemented for this exercise on two Arduino Due microcontroller boards. Sensor inputs are simulated by potentiometers (accelerometer, gyroscope, pressure sensor) and a digital switch (baby seat) on a breadboard. The output is a CAN bus frame verified at the OMS lab PC interface. The mechanical airbag module, inflator, and vehicle body structure are treated as adjacent systems in the Item Definition.

3. Assumptions and Boundary Conditions

3.1 General Assumptions

- The vehicle is a standard passenger car (M1 category) with a maximum design speed of 250 km/h.
- The Airbag ECU operates over the full vehicle lifetime, estimated at 15 years / 300,000 km.
- The ECU is powered from the vehicle 12V battery system with backup capacitor sufficient for ≥ 150 ms post-accident operation.
- The vehicle has between 1 and 8 occupants. In the laboratory setup, occupant classification is simplified: a single digital input (baby-seat switch) indicates whether the passenger seat is occupied by a child seat (LOW) or an adult / empty (HIGH). In a production vehicle, this would be provided by an Occupant Classification System (OCS) via CAN message.
- Ambient operating temperature range: -40°C to $+85^{\circ}\text{C}$ (storage: -40°C to $+105^{\circ}\text{C}$).
- EMC requirements are per CISPR 25 and ISO 11452 series.
- Squib firing energy is supplied by the ECU internal reservoir capacitor; squib resistance is within a known window ($1.5\ \Omega$ – $4.0\ \Omega$ nominal).

3.2 Operational Assumptions

- Normal operation: engine running, vehicle in motion, all sensors providing valid signals within specification.
- Crash detection algorithms process signals within 1 ms cycle time; deployment decision within 10–15 ms from impact onset.

- In the laboratory implementation, the ECU (Arduino Due) communicates via CAN bus to the OMS lab PC CAN interface. CAN frames carry the deployment decision bits, sensor status, and CRC field. A second parallel readback channel confirms the transmitted bitstream. In a production vehicle, CAN FD at 2 Mbit/s would connect the ECU to the BCM, instrument cluster, and OCS.
- Diagnostic Trouble Codes (DTCs) are logged in NVRAM and readable via OBD-II / UDS (ISO 14229).
- Self-tests are executed at every power-on cycle and continuously in the background (at least every 100 ms).
- A Diagnostic Warning Indicator (DWI) / airbag fault lamp is activated in the cluster upon detection of any system fault.

3.3 Environmental Assumptions

- Road surfaces: dry, wet, icy, gravel, pothole — all generating varying mechanical shock signatures.
- Vehicle use cases include: highway driving, urban driving, off-road, parking, towing, and wash/cleaning.
- Vibration environment: ISO 16750-3 road vehicle vibration profile.
- The ECU is mounted rigidly to the vehicle floor tunnel or firewall and is subject to the full crash pulse.

3a. Laboratory Hardware Setup

This section describes the concrete hardware implementation used in the Functional Safety laboratory exercise. The system architecture implements a two-channel (2oo2) redundant design as required by the ASIL D / SIL 3 classification derived in this HARA. All real vehicle sensors are substituted with laboratory equivalents to permit safe V&V testing.

3a.1 Sensor Inputs (Breadboard)

#	Real Sensor	Lab Substitute	Signal Type	Voltage Range	Represents
1	Accelerometer	POTI-1 (Potentiometer 1)	Analog (ADC)	0–3.3 V	0–100 g deceleration (crash pulse)
2	Gyroscope / Angular Rate	POTI-2 (Potentiometer 2)	Analog (ADC)	0–3.3 V	0–360°/s angular rate (rollover / yaw)

3	Pressure Sensor	POTI-3 (Potentiometer 3)	Analog (ADC)	0–3.3 V	0–500 kPa impact pressure
4	Baby Seat / OCS	Digital Switch (active-LOW)	Digital (GPIO)	LOW / HIGH (3.3 V)	LOW = baby seat present (suppress pax airbag); HIGH = adult / empty

3a.2 Processing (Dual-Channel Arduino Due)

Two Arduino Due boards implement the dual-channel (2oo2) architecture required for ASIL D. Each board independently reads all four sensor inputs and computes the deployment decision. A deployment output is only produced when both channels agree. The boards communicate their decisions for cross-comparison. Each board runs an identical embedded software image including CRC calculation using an Arduino CRC library.

3a.3 Output and Verification (CAN Bus → OMS Lab PC)

The deployment decision and sensor status are transmitted as a CAN bus frame from the Arduino Due to the CAN interface in the OMS lab, where a PC reads and logs the messages. A second parallel readback of the CAN bitstream detects transmission errors. CAN frame integrity is verified by the CRC field calculated on the Arduino using a standard CRC library. The failure probability of the CAN bus must remain within the limits derived from the ASIL D / SIL 3 requirement ($PMHF < 10^{-8} / h$). The test protocol records: scene settings (poti positions), expected CAN output, actual CAN output, and pass/fail result for each test case.

4. Use Cases

The following operating and use case scenarios are considered for hazard identification:

UC-ID	Use Case	Description / Relevance
UC-01	Normal Highway Driving	Vehicle traveling at 80–250 km/h on motorway. Highest crash severity potential. Full sensor fusion active.
UC-02	Urban / Low-Speed Driving	Vehicle at 0–50 km/h in urban environment. Pedestrian presence likely. Frequent speed changes; minor impacts common (door dings, parking).
UC-03	Frontal Crash (Barrier Impact)	Full-width or offset frontal impact. Primary scenario for frontal airbag and seatbelt pretensioner activation.

UC-04	Side Crash (Pole / MDB)	Lateral impact from another vehicle or rigid pole. Requires rapid (< 8 ms) side curtain and thorax airbag deployment.
UC-05	Rear Impact	Impact from rear. Seatbelt pretensioners activated; whiplash protection. Rear airbag systems where fitted.
UC-06	Rollover	Vehicle roll detected by yaw/roll rate sensors. Curtain airbags must remain deployed for 5+ seconds.
UC-07	Minor Non-Deployable Impacts	Pothole, speed bump, minor collision (< 15 g). No deployment must occur. System must distinguish from crash events.
UC-08	Vehicle Stationary / Ignition Off	Engine off, low-power standby mode. ECU remains powered by battery. Risk of inadvertent activation due to maintenance, theft, or battery voltage transients.
UC-09	Maintenance / Repair (Workshop)	Technician accessing vehicle interior; OBD-II connected; airbag system not fully isolated. Risk of inadvertent deployment during servicing.
UC-10	Vehicle Wash / Flood Wading	Vibration from wash equipment; immersion risk. Sensor noise must not trigger deployment. Ingress protection requirement.
UC-11	Child / Small Occupant Presence	OCS classifies occupant as child or empty seat. Passenger airbag must be suppressed to prevent injury from airbag deployment.
UC-12	End-of-Life / Scrapping	Airbag must be safely disarmed and deployed (controlled) before vehicle is scrapped. Pyrotechnic disposal risk if not handled correctly.

5. Normal Operation Description

5.1 Power-On Self-Test (POST)

Upon every ignition cycle, the Airbag ECU performs a complete self-test sequence within 800 ms:

- Microcontroller (MCU) CPU core and memory (RAM, Flash CRC) diagnostic — verified by a lockstep core or checkerboard test.
- Internal clock monitor and watchdog timer self-test.
- Squib loop resistance measurement for all channels (open-circuit, short-to-battery, short-to-ground detection).
- Sensor input plausibility check: all three potentiometer channels (POTI-1, POTI-2, POTI-3) read within valid ADC range (0–1023 on Arduino Due 12-bit ADC); values outside range flag a sensor fault.
- Power supply voltage monitoring (nominal: 12 V $\pm$ 2 V; backup capacitor charge verification).

- CAN bus communication verification: a test frame is sent from the Arduino Due to the OMS lab PC CAN interface; receipt is confirmed. CRC of the test frame is verified using the Arduino CRC library. Baby-seat digital input line-continuity check performed.

If any POST item fails, the corresponding DTC is stored and the airbag warning lamp is illuminated. The system remains in a safe degraded state.

5.2 Continuous Crash Monitoring

During normal vehicle operation the ECU continuously acquires and processes signals from:

- 1× accelerometer channel — simulated in the laboratory by Potentiometer 1 (POTI-1) on the breadboard. Voltage range 0–3.3 V represents 0–100 g equivalent. Read by both Arduino Due boards independently (Channel A and Channel B).
- 1× gyroscope / angular rate sensor channel — simulated in the laboratory by Potentiometer 2 (POTI-2) on the breadboard. Voltage range 0–3.3 V represents 0–360°/s angular rate equivalent. Read by both Arduino Due boards independently.
- 1× pressure sensor channel — simulated in the laboratory by Potentiometer 3 (POTI-3) on the breadboard. Voltage range 0–3.3 V represents 0–500 kPa equivalent. Read by both Arduino Due boards independently.
- 1× baby-seat digital input — implemented as a digital switch (active-LOW) on the breadboard connected to both Arduino Due boards. LOW = baby seat detected (suppress passenger airbag); HIGH = adult occupant or empty seat.

The crash algorithm on each Arduino Due independently reads all three sensor potentiometers (POTI-1 accelerometer, POTI-2 gyroscope, POTI-3 pressure) and evaluates whether readings exceed the defined deployment thresholds. A deployment decision is only authorised when: (1) the sensor readings on Channel A (Arduino 1) AND Channel B (Arduino 2) both independently exceed the threshold, (2) both channels agree on the decision (AND-gate dual-channel comparison), and (3) the baby-seat digital input is HIGH (no child seat). The CAN frame carrying the decision is transmitted with a CRC calculated using an Arduino CRC library. A second parallel readback verifies the transmitted bitstream on the CAN bus.

5.3 Deployment Actuation

Upon a valid deployment decision, the ECU:

- Opens the firing transistor gates for the relevant squib channels in the determined sequence.
- Applies regulated energy from the reservoir capacitor to the squib (typically 1.5–3.0 A for 1.8–3.0 ms).
- Records deployment event (timestamp, crash severity classification, occupant weight) in NVRAM.
- Broadcasts a crash notification frame on CAN (for fuel cutoff, door unlock, emergency call trigger).

Post-deployment: the ECU enters a latched fault state. Continued self-monitoring remains active. The airbag warning lamp is permanently illuminated. Reset requires authorised OBD workshop procedure.

6. Error Handling and Degradation Concept

Failure Mode	Detection Mechanism	Safe State / Reaction	Warning Indication
Sensor open circuit / short	Squib loop monitoring, resistance out of window	Affected squib channel disabled; DTC stored; remaining system continues	Airbag warning lamp ON; DTC logged
MCU internal fault (lockstep mismatch)	Dual lockstep core comparison every cycle	MCU hardware reset; watchdog re-engage; if recurring → system shutdown	Warning lamp ON; vehicle CAN fault frame
Power supply undervoltage (< 9 V)	Analog voltage supervisor circuit	ECU enters low-power hold; backup cap charged; deployment still possible	Warning lamp ON; cluster message
Power supply overvoltage (> 16 V)	Analog voltage clamp + interrupt	Firing transistors inhibited; capacitor charge halted; DTC stored	Warning lamp ON
CAN bus communication loss	CAN frame timeout (> 50 ms)	Deployment algorithm continues using last valid OCS weight class; pretensioners not suppressed	Warning lamp ON; limp-home mode
Flash memory CRC error	CRC32 check at power-on and periodic	Deployment inhibited; DTC stored; system shutdown	Warning lamp ON; cannot deploy
Excessive temperature (> 90 °C)	NTC temperature sensor on ECU PCB	Monitoring continues; if > 105 °C → deployment inhibited; DTC stored	Warning lamp ON
Satellite sensor LIN fault	LIN frame CRC / parity error > 3 consecutive	Affected satellite sensor excluded from algorithm	Warning lamp ON; DTC for affected sensor
Inadvertent clock deviation	Watchdog timer + PLL lock monitor	System reset via independent watchdog	Warning lamp ON

7. Identification of Hazardous Conditions

Hazardous conditions are identified through structured brainstorming using FMEA/HAZOP methods, considering each system function and possible failure mode (commission and omission failures):

7.1 Hazard Category H1 : Unintended Airbag Deployment

- Inadvertent deployment during normal driving — no crash event present. Could injure occupants, cause driver distraction and loss of vehicle control.
- Deployment due to EMC disturbance, software runaway, or spurious logic triggering the firing circuit.
- Deployment during vehicle wash, maintenance, or stationary operation — no occupant protection intended.
- Deployment in an inappropriate situation: wrong airbag (e.g., passenger airbag with child/empty seat in UC-11).

7.2 Hazard Category H2 : Non-Deployment in a Crash Requiring Deployment

- Failure to deploy any airbag in a frontal crash above threshold — severe injury or fatality risk to belted occupant.
- Failure to deploy side curtain in a lateral or rollover event — head injury from side structure contact.
- Failure to fire seatbelt pretensioners — occupant not restrained properly before airbag contact.
- Delayed deployment beyond 15 ms from crash threshold crossing — occupant already in contact with steering wheel or A-pillar.

7.3 Hazard Category H3 : Partial or Incorrect Deployment Sequence

- Deployment of incorrect airbag zone (e.g., rear instead of frontal) — no protection where needed.
- Deployment with incorrect timing between driver and passenger bags — wave interference reducing protection efficacy.
- Single squib ignition failure in a dual-stage inflator — under-inflation of bag.

7.4 Hazard Category H4 : Loss of Diagnostic Capability / Warning

- Permanent silence of DWI lamp despite system fault — occupant unaware of reduced protection.

- Loss of CAN broadcast after crash — fuel pump not cut, emergency call not triggered.

8. Baby Seat Digital Input : Hardware Failure Mode Analysis

The exercise hardware specifies a dedicated digital input line to the Arduino Due / ECU indicating whether the passenger seat is occupied by a baby seat. This input must be analysed for stuck-fault conditions, as either failure direction produces a distinct and serious safety consequence. The input is implemented as an active-LOW signal: LOW (0 V) = baby seat detected; HIGH (3.3 V / 5 V) = no baby seat / adult occupant.

Failure Mode	ECU Reads	Consequence	ASIL / SIL	Mitigation
Stuck HIGH (line open-circuit or pull-up fault)	HIGH → “No baby seat” (even though baby seat IS present)	CRITICAL. Passenger airbag NOT suppressed. In a crash, airbag deploys into baby seat. Likely fatal for infant. Equivalent to Hazard H1-04.	ASIL D / SIL 3	Suppress-on-fault default. Periodic line-continuity check (drive line LOW and verify). DWI lamp ON if continuity check fails.
Stuck LOW (short to ground)	LOW → “Baby seat present” (even with adult passenger)	SIGNIFICANT. Passenger airbag always suppressed. In a crash with an adult passenger, the passenger is unprotected by the frontal airbag; increased injury risk, though seatbelt pretensioner remains active. Maps to Hazard H2-01 (passenger).	ASIL C / SIL 2	DWI lamp ON and DTC logged immediately on detection of stuck-LOW. Occupant warned to visit workshop. The lower ASIL vs stuck-HIGH reflects that an adult has residual belt protection.
Intermittent / noisy signal	Undefined / toggling	Race condition may allow airbag to fire or be suppressed inconsistently. Treated as a fault condition.	ASIL D / SIL 3	Debounce filter (min. 10 ms stable before state change accepted). Any toggle faster than 10 ms triggers fault; suppress-on-fault applied. DWI lamp ON.

Note: The Arduino Due digital input for the baby seat must include a hardware pull-up resistor (10 kΩ to 3.3 V) so that an open-circuit wire defaults to HIGH. Combined with the suppress-on-fault (default-HIGH → suppress) software strategy, both the hardware pull-up and the software response together ensure that an open-circuit fault always results in passenger airbag suppression — the safest default.

9. ASIL/SIL Determination Parameters

9.1 Severity (S) : ISO 26262 Table B.1

Level	Description	Example
S0	No injuries	Purely economic damage, no harm to persons
S1	Light and moderate injuries	Reversible injuries; first-aid level
S2	Severe and life-threatening injuries (survival probable)	Broken bones, concussion, internal bleeding (survivable)
S3	Life-threatening injuries (survival uncertain) or fatal	Traumatic brain injury, thoracic trauma, fatality

9.2 Exposure (E) : ISO 26262 Table B.2

Level	Description	Frequency / Probability
E0	Incredibly unlikely	< once per vehicle lifetime
E1	Very low probability	A few times per vehicle lifetime (< 1%)
E2	Low probability	A few times per year (~1–10%)
E3	Medium probability	Fairly often (~10–50%)
E4	High probability	Almost always or very frequently (> 50%)

9.3 Controllability (C) : ISO 26262 Table B.3

Level	Description	Population Controllable
C0	Generally controllable	Simply avoidable by most drivers
C1	Simply controllable	90% of drivers can avoid harm
C2	Normally controllable	Most drivers (< 90%) can avoid; requires prompt reaction
C3	Difficult to control or uncontrollable	< 10% of drivers can avoid; situation evolves too fast

9.4 ASIL Lookup : ISO 26262 Part 3 Table 4

ASIL is determined from the combination of S, E, and C. The mapping below represents the standard ISO 26262 risk graph:

S \ E	E1	E2	E3	E4	C level →
S1	C1: QM / C2: QM / C3: QM	C1: QM / C2: QM / C3: ASIL A	C1: QM / C2: ASIL A / C3: ASIL B	C1: ASIL A / C2: ASIL B / C3: ASIL C	→ C1 / C2 / C3
S2	C1: QM / C2: QM / C3: ASIL A	C1: QM / C2: ASIL A / C3: ASIL B	C1: ASIL A / C2: ASIL B / C3: ASIL C	C1: ASIL B / C2: ASIL C / C3: ASIL D	→ C1 / C2 / C3
S3	C1: QM / C2: ASIL A / C3: ASIL B	C1: ASIL A / C2: ASIL B / C3: ASIL C	C1: ASIL B / C2: ASIL C / C3: ASIL D	C1: ASIL C / C2: ASIL D / C3: ASIL D	→ C1 / C2 / C3

Note: QM = Quality Management (no specific ASIL requirement). ASIL D is the highest safety integrity level.

9.5 SIL ↔ ASIL Equivalence Mapping (IEC 61508 / ISO 26262)

ASIL (ISO 26262)	SIL (IEC 61508)	PFH (Probability Failure per Hour)	Notes
QM	None / < SIL 1	—	No safety integrity requirement
ASIL A	SIL 1	$10^{-5} - 10^{-6} / h$	Low demand mode SIL 1
ASIL B	SIL 1–2	$10^{-6} - 10^{-7} / h$	Lower bound SIL 2
ASIL C	SIL 2–3	$10^{-7} - 10^{-8} / h$	Upper bound SIL 2 / lower SIL 3
ASIL D	SIL 3	$< 10^{-8} / h$	Highest automotive safety level

10. Hazardous Events, Safe States & ASIL/SIL Assignment

The table below presents the full HARA results. Severity (S), Exposure (E) and Controllability (C) ratings are per ISO 26262 Annex B. The ASIL and SIL are the maximum values determined across all contributing use cases.

ID	Hazard	Situation	Hazardous Event	S	E	C	SIL (IEC)	ASIL (ISO 26262)	Safe State	Remark
H1-01	Unintended deployment frontal airbag	Normal highway driving (UC-01); no crash event	Airbag deploys unexpectedly; driver blinded/struck; loss of vehicle control at speed	S3	E4	C3	SIL 3	ASIL D	No deployment without valid crash signal; firing inhibit logic active	Highest risk primary safety goal
H1-02	Unintended deployment side/curtain airbag	Urban driving (UC-02); no side impact	Curtain deploys unexpectedly; driver vision obstructed; loss of control	S3	E3	C3	SIL 3	ASIL D	Firing inhibit; dual-algorithm AND-gate required for side deployment	Also covers UC-07 (pothole misfire)
H1-03	Inadvertent deployment vehicle stationary, ignition off	Vehicle parked, maintenance (UC-08, UC-09); technician working in cabin	Airbag deploys and injures technician or bystander	S2	E2	C2	SIL 2	ASIL B	Firing transistor requires active MCU command; no stored energy path without MCU enable	Requires authorised disarm procedure in workshop
H1-04	Deployment of passenger airbag with child seat / empty seat	Child or empty seat present (UC-11); frontal crash occurs	Passenger airbag deploys against occupied child seat or into empty seat unnecessarily; risk of child injury	S3	E3	C3	SIL 3	ASIL D	OCS weight classification received and validated; passenger bag suppressed if OCS = child/empty; fallback = suppress if OCS CAN lost	OCS communication must also be ASIL-rated; suppress-on-fault strategy

H1-05	EMC-induced spurious deployment	Vehicle near high-power transmitter or experiencing severe ESD (any UC)	Electromagnetic disturbance causes firing transistor to open; airbag deploys	S3	E2	C3	SIL 2	ASIL C	Galvanic isolation on squib lines; EMC hardened firing circuit (differential signal, shielded squib wires); hardware firing key required	EMC tested per CISPR 25 / ISO 11452
H2-01	Non-deployment — frontal airbag in severe crash	Frontal barrier impact at > 30 km/h (UC-03); airbag algorithm fails to fire	Occupant unrestrained by airbag; head strikes steering wheel/windshield; severe injury or fatality	S3	E3	C3	SIL 3	ASIL D	ECU signals fault via DWI if system has detected self-test failure; system remains as armed as possible; no graceful degradation of deployment inhibit without fault	Primary safety goal — non-deployment = commission to omission failure
H2-02	Non-deployment — side curtain / thorax airbag in lateral crash	Pole impact at > 20 km/h (UC-04); side satellite sensor or algorithm fails	Occupant head contacts door trim/A-pillar; severe TBI risk	S3	E3	C3	SIL 3	ASIL D	Redundant satellite sensor signals; if primary sensor fails, secondary sensor path triggers deployment; DWI ON if sensor is permanently faulted	Deployment window: < 8 ms from impact threshold
H2-03	Non-deployment — seatbelt pretensioners in any crash	Any significant crash event (UC-03 to UC-06); pretensioner	Occupant not restrained before airbag contact; rib/clavicle fracture risk;	S2	E3	C3	SIL 2	ASIL C	Pretensioner squib loops monitored continuously; fault results in DWI ON but	Belt pretensioners considered secondary restraint; airbags remain primary

		circuit faulted	increased airbag injury potential						does not inhibit other channels	
H2-04	Delayed deployment (> 15 ms from threshold)	Frontal or side crash (UC-03, UC-04); algorithm processing delayed by MCU fault or watchdog reset	Occupant already moving forward at time of bag deployment; bag contacts occupant out-of-position; injury severity increased	S2	E2	C3	SIL 2	ASIL C	Lockstep MCU ensures algorithm cycle time ≤ 1 ms; hardware watchdog timeout < 10 ms; algorithm deadline monitored	Timing is safety-critical; must be specified as a safety goal
H3-01	Incorrect airbag zone deployment	Frontal crash (UC-03) but side airbag fires instead	No frontal protection; side airbag deploys unexpectedly causing injury	S3	E1	C3	SIL 2	ASIL B	Channel addressing verified by squib loop resistance matching; output address decoded in hardware; software channel map integrity checked at startup	Low exposure — requires specific HW wiring fault plus SW fault
H3-02	Single squib failure in dual-stage inflator	Severe frontal crash (UC-03); stage 2 squib circuit open	Bag under-inflated; insufficient restraint for occupant in high-speed crash	S2	E2	C2	SIL 1	ASIL A	Both squib resistances monitored independently; stage 1 deployment still occurs; DTC logged	Residual protection from stage 1; injury not fatal in most scenarios
H4-01	Loss of DWI / airbag warning lamp (silent fault)	Any operational mode; DWI circuit open; occupant unaware of system fault	Occupant drives with reduced airbag protection unaware; fatality risk in subsequent crash	S1	E3	C2	SIL 1	ASIL A	DWI lamp circuit verified at power-on (lamp drive current monitored); BCM cross-check via CAN	Driver can check lamp illumination at ignition-on; low ASIL relative to deployment hazards

H4-02	Loss of post-crash CAN broadcast	After deployment (any UC-03 to UC-06); CAN transceiver damaged by crash	Fuel pump not cut; emergency call (eCall/ERA-GLONASS) not triggered; fire risk or delayed emergency response	S2	E3	C2	SIL 1	ASIL B	CAN message transmitted immediately at deployment decision (before structural damage propagates); hardware watchdog ensures CAN TX attempt	Post-crash CAN survivability limited by vehicle structural integrity — beyond ECU control after severe crash
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11. Worked ASIL/SIL Determination Examples

The following step-by-step examples demonstrate the S/E/C rating process for two key scenarios specified in the exercise brief, following the same methodology as the high beam / low beam lecture example.

11.1.1 Example 1 Unintended Airbag Unfolding During Normal Driving

Hazard	Unintended deployment of the driver frontal airbag with no crash event present
Situation	Vehicle travelling at highway speed (80–250 km/h). No impact has occurred. The ECU erroneously issues a firing command due to a software runaway or EMC event.
Severity (S)	S3 — Life-threatening or fatal. The airbag strikes the driver in the face/chest at high speed while the vehicle is in motion; driver loses control; risk of multi-vehicle collision and fatality.
Exposure (E)	E4 — High probability. The vehicle is driven regularly; the ECU is powered and monitoring continuously during every trip. The operational situation (driving) occurs almost every day throughout the vehicle's 15-year lifetime.
Controllability (C)	C3 — Difficult or impossible to control. The deployment is instantaneous (~3 ms inflation). The driver has no warning and no ability to react. Less than 10% of drivers could avoid subsequent loss of control.
Result	S3 + E4 + C3 → ASIL D / SIL 3 (Safety Goal SG-01)

Required Measure	Dual-algorithm AND-gate deployment logic; hardware firing inhibit path independent of software; dual lockstep MCU; ASIL D SW/HW development per ISO 26262 Parts 4 & 6.
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11.1.2 Example 2 Airbag Deploys with Baby Seat / Child Seat Installed

Hazard	Passenger airbag deploys into an installed rear-facing baby seat or forward-facing child seat during a crash event
Situation	A baby/child seat is installed on the front passenger seat. A frontal crash occurs above the deployment threshold. The ECU has not correctly read or acted on the baby-seat digital input signal, and the passenger airbag fires.
Severity (S)	S3 — Life-threatening or fatal. An airbag deploying directly into a rear-facing child seat exerts a force far exceeding a child's neck/head tolerance. The injury is likely fatal for infants.
Exposure (E)	E3 — Medium probability. Child/baby seats are commonly used; statistics show a significant proportion of family vehicles carry child seats regularly. The situation occurs fairly often across the vehicle population.
Controllability (C)	C3 — Difficult or impossible to control. The deployment happens in under 30 ms total. No occupant or driver action can prevent the airbag from striking the child seat once triggered. The failure is entirely within the ECU.
Result	S3 + E3 + C3 → ASIL D / SIL 3 (Safety Goal SG-03)
Required Measure	Dedicated baby-seat digital input (active-low signal from seat clip sensor) read every 1 ms cycle. Suppress-on-fault strategy: if input is unreadable or signal lost, passenger airbag is inhibited by default. See Section 16 for digital input failure modes.

12. Safe States Definition

ISO 26262 requires that each safety goal be assigned a safe state — the condition the system must achieve or maintain when a fault is detected. The following safe states are applicable to the Airbag ECU:

Safe State ID	Description	Applicable Hazard(s)	System Condition
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SS-01	No deployment (inhibit state)	H1-01, H1-02, H1-03, H1-05	All firing transistors remain open (off). Reservation capacitor maintained but no firing command issued. DWI lamp ON if fault detected.
SS-02	Deployment armed, awaiting valid crash signal	H2-01, H2-02, H2-03	Normal standby. All squib channels armed. Algorithm running. Any fault disables only affected channel, not full system.
SS-03	Passenger airbag suppressed (OCS fault)	H1-04	If OCS CAN signal lost > 50 ms, passenger airbag is inhibited by default (suppress-on-fault). Driver and side airbags remain armed.
SS-04	Affected channel isolated (partial degradation)	H3-01, H3-02, H4-01	Faulted squib channel disabled; remaining channels continue normal operation. DWI lamp ON. System logs DTC.
SS-05	System shutdown (unrecoverable MCU fault)	H1-01, H2-01 (related)	All output transistors forced off by hardware safety switch. Watchdog asserts reset signal. If fault persists after 3 resets → permanent shutdown + DWI lamp.

13. Summary of ASIL / SIL Determination

The following table summarises the final safety integrity levels per safety goal. The highest ASIL derived from the HARA governs the functional safety requirements for each item function:

Safety Goal ID	Safety Goal	ASIL (ISO 26262)	SIL (IEC 61508)	Governing Hazard(s)
SG-01	The Airbag ECU shall not command deployment of any airbag or pretensioner in the absence of a valid crash event.	ASIL D	SIL 3	H1-01, H1-02, H1-04
SG-02	The Airbag ECU shall command deployment of the appropriate airbag(s) within the required time window when a crash event exceeding the deployment threshold is detected.	ASIL D	SIL 3	H2-01, H2-02
SG-03	The Airbag ECU shall suppress the passenger airbag when the OCS indicates a child occupant or empty seat.	ASIL D	SIL 3	H1-04
SG-04	The Airbag ECU shall command seatbelt pretensioner activation within 5 ms of a deployment decision.	ASIL C	SIL 2	H2-03, H2-04
SG-05	The Airbag ECU shall not command deployment in response to an EMC or electrical disturbance that does not represent a crash event.	ASIL C	SIL 2	H1-05
SG-06	The Airbag ECU shall activate the diagnostic warning indicator (DWI/airbag lamp) within 800 ms of detecting a system fault.	ASIL A	SIL 1	H4-01
SG-07	The Airbag ECU shall transmit a crash notification CAN frame	ASIL B	SIL 2	H4-02

	within 15 ms of a deployment decision.			
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Overall Item ASIL: ASIL D (ISO 26262) | Overall Item SIL: SIL 3 (IEC 61508)

The Airbag ECU item is classified ASIL D. All hardware and software components realising Safety Goals SG-01, SG-02, and SG-03 must be developed in accordance with ISO 26262 Part 4 (hardware) and Part 6 (software) ASIL D requirements, including:

- Dual-core lockstep MCU architecture with hardware safety monitor.
- Hardware-enforced firing inhibit path independent of software (hardware safety switch / e-fuse).
- 100% MC/DC code coverage for safety-relevant software modules.
- FMEA with single-point fault metric (SPFM) ≥ 99% for ASIL D functions.
- Probabilistic metric for hardware random failures (PMHF) < 10 FIT for ASIL D safety goals.

14. References and Normative Standards

- ISO 26262:2018 — Road Vehicles — Functional Safety (Parts 1–12)
- IEC 61508:2010 — Functional Safety of Electrical/Electronic/Programmable Electronic Safety-related Systems (Parts 1–7)
- ISO 21448:2022 — SOTIF (Safety of the Intended Functionality)
- ISO 16750-3 — Road Vehicles — Environmental Conditions and Testing — Mechanical Loads
- CISPR 25:2016 — Vehicles, Boats and Internal Combustion Engines — Radio Disturbance Characteristics
- ISO 11452 — Road Vehicles — Component Test Methods for Electrical Disturbances from Narrowband Radiated Electromagnetic Energy
- ISO 14229-1:2020 — Unified Diagnostic Services (UDS)
- SAE J211 — Instrumentation for Impact Tests
- Federal Motor Vehicle Safety Standard (FMVSS) No. 208 — Occupant Crash Protection
- UN Regulation No. 94/95 — Frontal / Side Collision

15. Revision History

Rev	Date	Author	Change Description
1.0	2026-05-10	Safety Engineering Team	Initial release - full HARA covering all 13 identified hazardous events; 7 safety goals; ASIL D / SIL 3 classification confirmed.
2.0	2026-05-10	Safety Engineering Team	Rev 2.1 — Updated to reflect confirmed lab hardware: 2 Arduino Due boards (dual-channel architecture), 3 potentiometers (POTI-1 accelerometer, POTI-2 gyroscope, POTI-3 pressure sensor) on breadboard, 1 digital baby-seat switch. Section 2 scope, Section 3.1/3.2 assumptions, Section 5.1 POST, and Section 5.2 sensor list updated accordingly. CAN bus output to OMS lab PC and parallel bitstream readback documented.

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